Exploring Electricity with PhET Circuit Construction Kit (DC)

Today's Lesson Focus

Students will transition from theoretical understanding to practical application, actively building, experimenting with, and measuring DC circuits using the interactive PhET Circuit Construction Kit simulation.

Estimated Time Needed

Approximately 45-60 minutes. This can be adjusted based on the depth of exploration and pre-existing student knowledge of electricity concepts.

Materials & Setup Required

- Computers or tablets with reliable internet access
- Access to the PhET DC Circuit Construction Kit (DC) simulation (pre-load for efficiency)
- Optional: Student worksheet or guided inquiry questions for structured experimentation

Key Learning Outcomes

- Students will successfully construct and manipulate various basic DC circuits (series and parallel).
- Students will predict and measure current and voltage relationships within circuits.
- Students will explain the function of common circuit components (batteries, wires, light bulbs, resistors).
- Students will develop an intuitive understanding of Ohm's Law through observation and experimentation.

Teacher's Transition Script: Recap & Hands-On Circuits

Teacher Says:

Alright everyone, settle in! Last time we met, we had a blast exploring the exciting world of electricity with Bill Nye and those really clear BBC videos. We talked about some big ideas that are fundamental to how everything from your phone to our classroom lights work. Can anyone remind me of one or two key things we learned about electricity?

Class Does:

Students raise hands, offering answers like: "how current flows," "what voltage is," "conductors and insulators," "what makes a simple circuit work."

Teacher Says:

Fantastic! I heard "current" – that's the flow of electricity, right? And "voltage" – that's the 'push' that makes the current move. We also touched on conductors, like metals, which let electricity pass through, and insulators, like plastic or rubber, which block it. We even saw how to build a basic circuit with a battery, wires, and a light bulb.

Now, those videos gave us a great foundation. We *saw* the concepts, but how do we move from just watching to actually *doing*? How can we test these ideas and really understand them without having to buy a whole bunch of expensive equipment or risk a minor shock?

Class Does:

Students ponder, perhaps suggesting building things or experimenting, but looking for the solution.

Teacher Says:

Exactly! That's where our amazing digital tool comes in today: the PhET Circuit Construction Kit. Think of it as our virtual science lab where we can safely build, experiment with, and measure all sorts of circuits. We'll be taking those concepts of current, voltage, and circuit components that we learned about with Bill Nye and applying them directly. You'll be building your own circuits, seeing the electrons flow, and even testing out Ohm's Law firsthand!

Class Does:

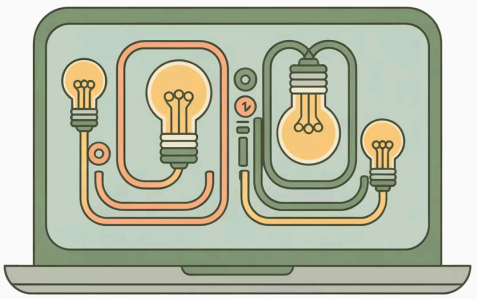
Students open their laptops or tablets, ready to access the PhET simulation.

Introduction to the PhET Circuit Construction Kit (DC)

What Is PhET?

PhET is a free, research-based interactive simulation developed by the University of Colorado Boulder. The DC Circuit Construction Kit allows students to build and test real-feeling circuits entirely in a browser — no physical materials needed.

- Build with batteries, wires, switches, bulbs, and resistors
- See electric current as animated "blue balls" (moving charges)
- Works on any modern browser — laptop, tablet, or Chromebook
- Ideal for in-person, hybrid, or fully remote classrooms



Making abstract electricity concepts **visible and tangible** — students literally watch electrons move.

Hey Scientists! Let's Connect the Dots!

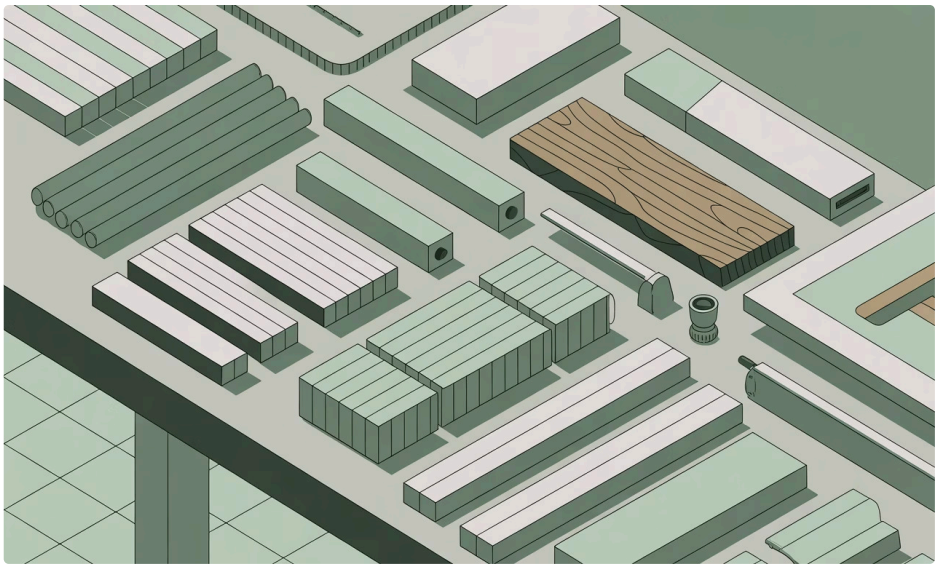
Remember all those cool things you learned from Bill Nye and the BBC? Well, guess what? The PhET simulator isn't just a game; it's where all those awesome science concepts come to life! Everything you watched, you can now control and explore yourself. Let's see how our simulator brings those video lessons right to your fingertips!

Teacher Says:

"First up, remember when Bill Nye showed us how opening a light switch breaks a circuit and turns off the lights? He made it look so simple! Now, in the PhET simulator, it's YOUR turn to be the circuit master! You can literally open and close switches, instantly seeing the light bulb turn on or off, and the electron flow start or stop. It's like having Bill Nye's demonstration right there on your screen, and YOU'RE in charge!"

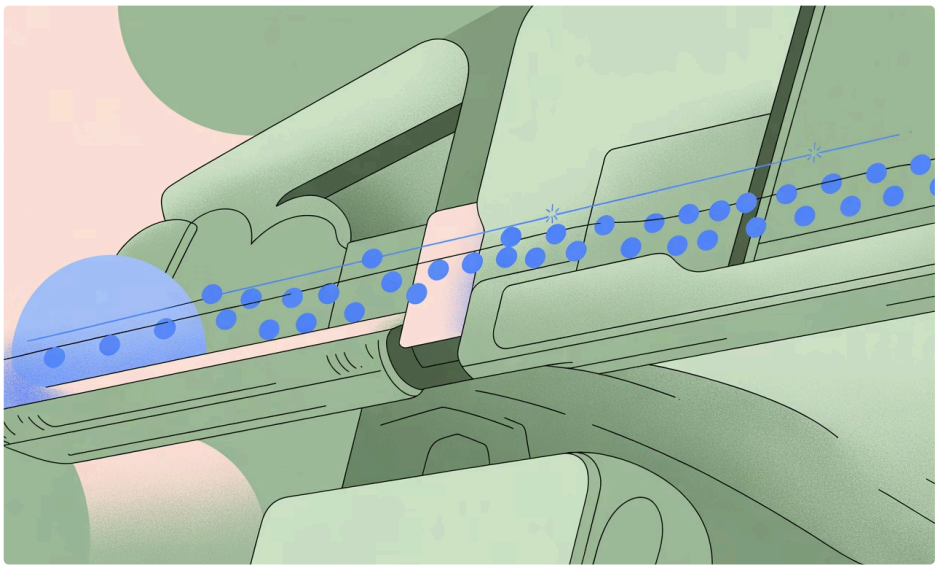
Teacher Says:

"Next, recall how the BBC explained why some materials, like metals, are amazing at letting electricity zoom through them (conductors), while others, like rubber or wood, block it completely (insulators). In the simulator, you don't just hear about it—you test it! Drag different items like paper clips or pencils into your circuit. Watch the light bulb: does it light up or stay dark? It's your chance to prove exactly what makes a conductor a conductor, and an insulator an insulator!"



Teacher Says:

"And probably one of the coolest parts: remember Bill Nye talking about electricity as tiny little bits flowing through wires? The PhET simulator makes those invisible bits VISIBLE! Those little blue balls aren't just for show—they're the electrons, and you can literally watch them move! See how they rush through conductors and get stuck at insulators? It's like a super-powered microscope showing you the electric current in action, just as Bill Nye described it!"



Teacher Says:

"So, the PhET simulator isn't just a new activity. It's your own personal science lab where you get to experiment, control, and prove everything you've been learning. Get ready to put on your scientist hats and make those awesome video lessons come alive!"

Learning Objectives for This Lesson

By the end of this session, students will be able to demonstrate the following skills and conceptual understandings through virtual experimentation in the PhET simulator.

1

Open vs. Closed Circuits

Understand and differentiate how current flows — or stops — depending on circuit state.

2

Conductors & Insulators

Identify which materials allow electrical flow through virtual "grab bag" testing.

3

Voltage & Current Measurement

Use virtual voltmeters and ammeters to take and interpret circuit readings.

4

Ohm's Law Exploration

Manipulate voltage and resistance to observe how current responds in real time.

5

Series & Parallel Circuits

Construct both circuit types and compare their electrical behaviors and properties.

Detailed Teaching Guide: Activity 1 – Building a Simple Closed Circuit

This guide provides step-by-step instructions for leading students through Activity 1 in the PhET Circuit Construction Kit: DC simulator. The goal is for students to understand the basic requirements for current flow.

Teacher Says:

"Welcome, electrical engineers! Today, we're going to start building our understanding of electricity by creating our very first circuit in the PhET simulator. Our first goal is to make a simple circuit that will light up a light bulb. First, please open the 'Circuit Construction Kit: DC' simulator. You'll see a blank workspace and a toolbox of components on the right."

Class Does:

- Open the PhET "Circuit Construction Kit: DC" simulator.
- Familiarize themselves with the workspace and component toolbox.

Teacher Says:

"Great! Your task for this activity is to build a simple closed circuit. Remember, a closed circuit is like a complete loop that allows electricity to flow. You will need three main components: a battery, a light bulb, and a switch. You'll also need wires to connect them. Drag these components onto your workspace now."

Class Does:

- Drag a **Battery** from the component toolbox to the workspace.
- Drag a **Light Bulb** to the workspace.
- Drag a **Switch** to the workspace.

Teacher Says:

"Now, think about how you connect them so that the electricity can travel from one end of the battery, through the bulb, through the switch, and back to the other end of the battery. Use the wires to create a continuous loop. Make sure all connections are secure – the ends of the wires should snap together with the component terminals. Take your time, and let me know if you run into any 'gaps'!"

Class Does:

- Drag **Wires** to connect the battery, bulb, and switch in a continuous loop.
- Ensure all connections are secure, forming a complete path.

Teacher Says:

"Alright, engineers, once you've connected everything, it's time for the moment of truth! Locate your switch and click it to **close the circuit**. Observe what happens to the light bulb. Also, pay close attention to the small blue dots moving in the wires – what do you notice about them?"

Class Does:

- Click the switch to **close the circuit**.
- Observe the flow of electrons (moving blue dots) and the brightness of the light bulb.

Teacher Says:

"Fantastic! Now, click the switch again to **open the circuit**. What changes this time? What happens to the light bulb, and what about those blue dots?"

Class Does:

- Click the switch again to **open the circuit**.
- Observe the flow of electrons and the light bulb in this new state.

Teacher Says:

"Excellent observations, everyone! Let's discuss what we just saw. When the switch was closed, what happened to the bulb and the blue dots? And then, what happened when you opened the switch? What do you think those small blue dots moving through the wires represent?"

Class Does:

- Share observations about the bulb lighting up/turning off.
- Describe the movement (or lack thereof) of the blue dots.
- Propose ideas about what the blue dots might represent (electrons/current).

Teacher Says:

"You're on the right track! Those blue dots are indeed our electrical current, the flow of electrons. Based on your experiments, what were the essential components needed to make the light bulb light up? And what happens if any part of your circuit is disconnected or if you don't have a complete loop? In your own words, how would you describe the difference between an 'open circuit' and a 'closed circuit' now?"

Class Does:

- Identify the battery, bulb, switch, and wires as essential components.
- Explain the necessity of a complete circuit for current flow.
- Define "open circuit" and "closed circuit" based on their observations and understanding.

📓 My Notes: Activity 1 — Simple Closed Circuit

Copy these notes into your science notebook!

Key Vocabulary

Circuit

A complete, closed path through which electric current can flow.

Open Circuit

A circuit with a break in the path — electricity cannot flow and the bulb will NOT light.

Closed Circuit

A complete, unbroken path — electricity CAN flow and the bulb WILL light.

Important Concepts to Copy

1. Electricity needs a CLOSED loop to flow. If there is any gap or break, the circuit is open and current stops.

2. A basic circuit has three parts: a power source (battery), a conductor (wire), and a load (light bulb).

3. When you connect all parts with no gaps, electrons flow from the battery through the wire to the bulb and back — this lights the bulb!

✏️ **Draw It!**

In your notebook, draw a simple closed circuit with: a battery, two wires, and a light bulb. Label each part.

My Circuit Drawing

Detailed Teaching Guide: Activity 2 – Testing Conductors and Insulators

In this activity, students will use the PhET Circuit Construction Kit: DC simulator to investigate which materials allow electricity to flow (conductors) and which do not (insulators). This builds on their understanding of closed circuits from Activity 1.

Teacher Says:

"Welcome back, budding engineers! In our last activity, we learned about closed circuits and how they allow electricity to flow and light up a bulb. Today, we're going to explore what kind of materials allow that flow, and what kind block it. We're talking about **conductors** and **insulators**."

Class Does:

- Listen attentively and prepare for the next activity.

Teacher Says:

"Your goal for this activity is to test different materials to determine if they are conductors or insulators. To do this, you'll need to modify the simple circuit you built last time. Disconnect one of the wires in your circuit – just create a gap. On the right side of your simulator, you should see a 'Grab Bag' filled with various objects like paper clips, coins, erasers, and pencils. This is where we'll find our test materials."

Class Does:

- Open the PhET "Circuit Construction Kit: DC" simulator.
- Disconnect one wire in their existing closed circuit to create a gap.
- Locate the "Grab Bag" section in the component toolbox.

Teacher Says:

"Let's start our investigation with a common metal object. From the 'Grab Bag,' drag a **Paper Clip** and place it in the gap you created in your circuit. Connect the loose ends of your wires to either side of the paper clip, ensuring a secure connection. Once connected, close your circuit switch and observe what happens to the light bulb and the blue dots."

Class Does:

- Drag a **Paper Clip** from the "Grab Bag" and connect it into the circuit gap.
- Close the circuit switch.
- Observe the light bulb and the movement of electrons (blue dots).

Teacher Says:

"What did you observe? Did the light bulb light up? Did the blue dots flow through the paper clip? Based on this, would you classify the paper clip as a conductor or an insulator, and why?"

Class Does:

- Describe observations (e.g., "The bulb lit up and the blue dots flowed").
- Conclude that the paper clip is a conductor because it allowed electricity to pass through.

Teacher Says:

"Excellent! Now, let's try a different material. Remove the paper clip from your circuit. This time, select a **Pencil** from the 'Grab Bag' and place it in the circuit gap. Connect your wires to it. Make sure you connect to the wood part of the pencil first. Close the switch and tell me what you see."

Class Does:

- Remove the **Paper Clip** from the circuit.
- Drag a **Pencil** from the "Grab Bag" and connect it into the circuit gap, ensuring connections to the wooden casing.
- Close the circuit switch and observe.

Teacher Says:

"Interesting! What happened this time? Did the bulb light up? Were the blue dots able to move through the wooden part of the pencil? What does this imply about wood – is it a conductor or an insulator?"

Class Does:

- Report observations (e.g., "The bulb did not light up, and the blue dots stopped at the pencil").
- Conclude that wood is an insulator because it blocked the flow of electricity.

Teacher Says:

"Fantastic observations! Now, let's explore the rest of the items in your 'Grab Bag.' I want each of you to systematically test at least three more different materials. For each material, predict whether it will be a conductor or an insulator before you test it, and then note your observations. Pay close attention to materials like the coin, the eraser, and even the graphite in the pencil if you connect to its tip. We'll discuss your findings in a few minutes."

Class Does:

- Independently select and test at least three additional materials from the "Grab Bag."
- Predict the material's property (conductor/insulator) before testing.
- Observe and record whether the bulb lights up and if electrons flow.

Teacher Says:

"Great work, everyone! Now, let's share what you found. Which materials were conductors, and which were insulators? What general properties do you notice about conductors compared to insulators? Think about what these materials are made of."

Class Does:

- Share their findings for the additional materials tested.
- Discuss common characteristics of conductors (e.g., metals) and insulators (e.g., rubber, plastic, wood).
- Formulate a basic understanding of why certain materials conduct electricity and others don't.

📓 My Notes: Activity 2 — Conductors & Insulators

Copy these notes into your science notebook!

Key Vocabulary

Conductor

A material that allows electricity to flow through it easily. Examples: copper wire, aluminum foil, pencil graphite.

Insulator

A material that does NOT allow electricity to flow through it. Examples: rubber, plastic, wood, glass.

Electric Current

The flow of electrons through a conductor.

Important Concepts to Copy

> Conductors let electrons move freely — they complete the circuit and allow the bulb to light up.

> Insulators block the flow of electrons — they break the circuit and the bulb will NOT light.

> Metals (like copper and aluminum) are almost always conductors. Non-metals (like rubber and plastic) are almost always insulators.

> Graphite (the "lead" in a pencil) is a non-metal that IS a conductor — it's a special exception!

✏️ **Draw It!**

In your notebook, draw a T-chart with two columns. Label one side 'Conductors' and the other 'Insulators.' List at least 3 examples in each column.

My T-Chart

Detailed Teaching Guide: Activity 3 – Measuring Voltage and Current with Resistors

In this activity, students will move beyond qualitative observations to quantitatively measure voltage and current in circuits, and begin to understand the role of resistors. This activity directly addresses learning objectives for voltage and current measurement, and introduces the foundational concepts of Ohm's Law.

Teacher Says:

"Hello again, aspiring electricians! So far, we've explored how circuits work and which materials conduct electricity. Now, we're going to dive deeper and learn *how much* electricity is flowing and *how much energy* it carries. Today, we'll introduce two crucial measurements: **voltage** and **current**, and then see how **resistors** change things. We'll be using two new tools in our simulator: the **voltmeter** and the **ammeter**."

Class Does:

- Listen attentively and prepare to learn about voltage, current, and resistors.

Teacher Says:

"First, let's get our basic circuit back. Build a simple closed circuit with a battery, a switch, and a light bulb. Now, find the **Voltmeter** in your toolkit – it looks like a circle with a 'V' inside. A voltmeter measures **voltage**, which is the electrical 'pressure' or energy difference between two points. To measure voltage, you connect its two probes **in parallel** across the component you want to measure. Let's start by measuring the voltage across your battery. Touch one probe to one terminal of the battery and the other probe to the other terminal."

Class Does:

- Assemble a simple circuit: battery, switch, light bulb.
- Locate the voltmeter in the PhET simulator toolkit.
- Connect the voltmeter probes across the battery and observe the reading.

Teacher Says:

"What reading do you get for your battery's voltage? It should be around 9 volts for a standard battery. Now, without disconnecting the voltmeter from the battery, connect its probes across the light bulb instead. What do you notice about the voltage reading across the bulb compared to the battery?"

Class Does:

- Record the battery voltage.
- Move the voltmeter probes to measure the voltage across the light bulb.
- Compare the two voltage readings.

Teacher Says:

"Excellent! You should see that the voltage across the bulb should be nearly the same as the battery's voltage, as long as the bulb is the only component in the circuit.. Now, let's measure **current**. Current is the flow rate of electrons through the circuit. Find the **Ammeter** – it looks like a circle with an 'A' inside. Unlike the voltmeter, an ammeter must be connected **in series** with the component whose current you want to measure. This means you have to break your circuit and insert the ammeter directly into the path of the current. Break the wire anywhere in your circuit and place the ammeter there. Close the switch and see what current flows."

Class Does:

- Locate the ammeter in the simulator toolkit.
- Break the circuit (e.g., disconnect a wire) and insert the ammeter into the gap.
- Close the switch and observe the current reading.

Teacher Says:

"Great! You now have a reading for the current in your circuit. Keep that number in mind. Now, let's introduce a new component: a **resistor**. Resistors are components designed to **resist** the flow of electricity, effectively reducing the current. Replace your light bulb with a resistor from the 'Grab Bag.' Start with a resistor value of about 10 ohms (you can adjust its value by clicking on it). Measure the voltage across this resistor and the current flowing through it. What changes do you observe in both the voltage and the current compared to when you had the light bulb?"

Class Does:

- Remove the light bulb and replace it with a resistor (e.g., 10 Ω).
- Use the voltmeter to measure voltage across the resistor.
- Use the ammeter to measure current through the resistor.
- Note changes in voltage and current readings.

Teacher Says:

"Fantastic! You should notice that the current has changed. Now, here's your challenge: try at least three different resistor values. For each value, record the resistance, the voltage across it, and the current through it. Look closely at your data – can you spot a pattern or a relationship between voltage, current, and resistance? This relationship is one of the most fundamental laws in electricity!"

Class Does:

- Select three different resistor values from the 'Grab Bag' or adjust the existing resistor's value.
- For each resistor, measure and record its resistance, the voltage across it, and the current through it. A small table might be useful here!

Resistance (Ω)	Voltage (V)	Current (A)
10	...	...
50	...	...
100	...	...

Teacher Says:

"Excellent work with the measurements! What did you discover? When you increased the resistance, what happened to the current? What about the voltage? Can you describe the mathematical relationship between these three quantities? Hint: try dividing voltage by current for each of your data points."

Class Does:

- Share their findings for different resistor values.
- Discuss how increasing resistance impacts current (current decreases) and voltage (voltage across the resistor should remain close to battery voltage if it's the only component).
- Calculate V/I for their data points and recognize that it approximates the resistance, introducing them to Ohm's Law ($V = IR$).

📓 My Notes: Activity 3 — Voltage, Current & Resistance

Copy these notes into your science notebook!

Key Vocabulary

Voltage (V)

The "push" or force that drives electrons through a circuit. Measured in Volts (V). The battery provides voltage.

Current (I)

The flow of electrons through a circuit. Measured in Amperes, or Amps (A).

Resistance (R)

How much a material opposes or slows down the flow of current. Measured in Ohms (Ω).

The Big Formula — Ohm's Law

> "V = I × R"
>
> "Voltage = Current × Resistance"
>
> This means: the more resistance in a circuit, the less current will flow (if voltage stays the same).

Important Concepts to Copy

- A voltmeter measures voltage. It is connected in PARALLEL (across) a component.
- An ammeter measures current. It is connected in SERIES (in line) with a component.
- Adding more resistance (a bigger resistor) to a circuit causes the current to DECREASE.
- Increasing the voltage (a bigger battery) causes the current to INCREASE.

✏️ Draw It!

In your notebook, write Ohm's Law (V = I × R) in large letters and draw a triangle divided into three sections. Write V on top, I on the bottom left, and R on the bottom right. This is the Ohm's Law triangle — use it to solve for any missing value!

My Ohm's Law Triangle

Detailed Teaching Guide: Activity 4 – Series vs. Parallel Circuits

This activity introduces students to the fundamental differences between series and parallel circuits, covering how voltage and current behave in each configuration and the practical implications for light bulb brightness.

Teacher Says:

"Welcome back, future engineers! Today we're moving beyond simple single-component circuits to explore how multiple components interact. Specifically, we're going to build circuits with several light bulbs and see how connecting them in **series** versus **parallel** dramatically changes their behavior. This is crucial for understanding everything from holiday lights to household wiring."

Class Does:

- Listen attentively and prepare to explore series and parallel circuits.

Teacher Says:

"Before we dive into building, think about holiday lights. Have you ever had a string of old Christmas lights where if one bulb burned out, the whole string went dark? That's an example of a series circuit. But with newer lights, if one bulb goes out, the rest often stay lit. That's a parallel circuit! Today, we're going to understand why that happens and how these different connections affect how electricity flows and how bright our bulbs shine."

Class Does:

- Reflect on real-world examples like Christmas lights and consider why they behave differently.
- Discuss initial thoughts on how one broken bulb could affect others in a circuit.

Teacher Says:

"Let's start with a **series circuit**. Remember how an ammeter needs to be connected **in series**? Think of it like a chain. In a series circuit, all components are connected end-to-end, forming a single path for the current. Build a new circuit with a battery, a switch, and **three** light bulbs connected one after another, all in a single loop. What do you observe about the brightness of the bulbs?"

Class Does:

- Assemble a series circuit with a battery, switch, and three light bulbs.
- Close the switch and observe the brightness of the bulbs.
- Note any differences compared to a single-bulb circuit.

Teacher Says:

"Now, let's take some measurements. Using your ammeter, measure the current at three different points in your series circuit: right after the battery, between the first and second bulb, and after the third bulb. What do you notice about the current readings at these different points?"

Class Does:

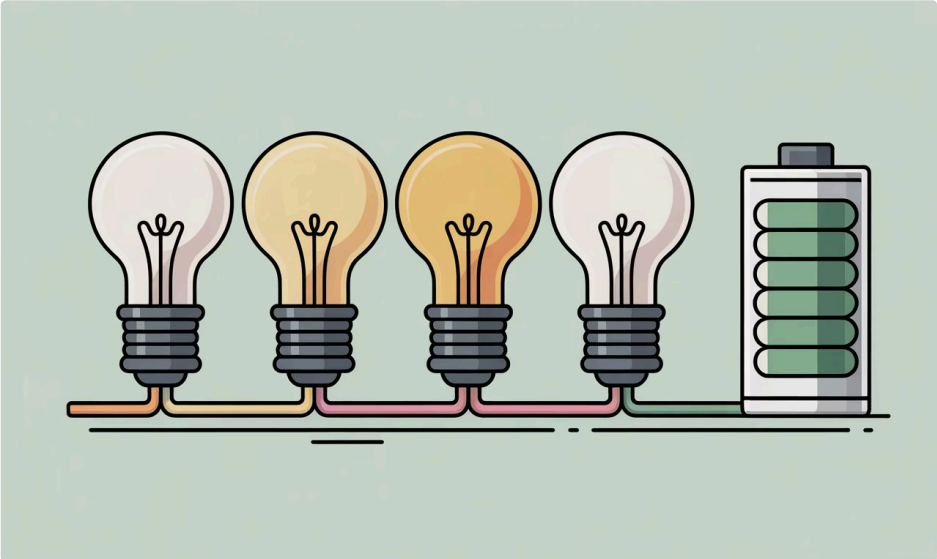
- Insert the ammeter at three distinct points in the series circuit.
- Record the current readings for each point.
- Observe that the current is the same throughout a series circuit.

Teacher Says:

"Excellent! You should find that the current is the same everywhere in a series circuit. Now, use your voltmeter to measure the voltage across **each individual light bulb**, and then measure the total voltage across **all three bulbs combined** (from the beginning of the first to the end of the last). How do the individual bulb voltages relate to the total voltage of the battery?"

Class Does:

- Measure the voltage across each of the three light bulbs individually.
- Measure the total voltage across the combination of the three bulbs.
- Record observations and compare individual voltages to the total (they should sum up to the battery's voltage).



Teacher Says:

"Alright, let's switch gears to a **parallel circuit**. In a parallel circuit, components are connected across the same two points, creating multiple independent paths for the current. This is like a river splitting into several streams and then rejoining. Disassemble your series circuit, and now build a parallel circuit with the same battery, switch, and **three** light bulbs. Each bulb should have its own loop directly connected to the battery. What do you observe about their brightness now, compared to the series circuit?"

Class Does:

- Assemble a parallel circuit with a battery, switch, and three light bulbs, each in its own parallel branch.
- Close the switch and observe the brightness of the bulbs.
- Note that the bulbs are much brighter than in the series circuit.

Teacher Says:

"Notice how much brighter they are! Now, let's measure voltages. Use your voltmeter to measure the voltage across **each individual light bulb** in your parallel circuit. What do you observe about these readings compared to the battery voltage?"

Class Does:

- Measure the voltage across each of the three light bulbs individually in the parallel circuit.
- Record observations and compare these voltages to the battery's voltage (they should all be approximately equal).

Teacher Says:

"Fantastic! You should see that the voltage across each bulb in a parallel circuit is approximately the same as the battery's voltage. Finally, let's measure current. Use your ammeter to measure the current flowing through **each individual branch** (i.e., through each bulb). Then, measure the total current coming out of the battery. How do the branch currents relate to the total current?"

Class Does:

- Measure the current through each of the three individual light bulb branches.
- Measure the total current flowing out of the battery.
- Record observations and compare individual branch currents to the total (individual currents should sum up to the total current).

Teacher Says:

"Excellent work with these observations and measurements! You've just uncovered the core differences between series and parallel circuits. Let's summarize our findings in a table to really cement this knowledge. Think about brightness, current, and voltage distribution, and what happens if one bulb breaks."

Class Does:

- Discuss their findings and prepare to compare the two circuit types.
- Fill in the table below to summarize the characteristics of series and parallel circuits.

Characteristic	Series Circuit	Parallel Circuit
Brightness of Bulbs	Dim, gets dimmer with more bulbs	Bright, brightness stays constant with more bulbs (up to a point)
Current (A)	Same everywhere in the circuit	Divides among branches; total current is sum of branch currents
Voltage (V)	Divides among components; sum of individual voltages equals total voltage	Same across each parallel branch (equal to total voltage)
Effect of Removing a Bulb	All other bulbs go out (circuit breaks)	Other bulbs remain lit (independent paths)

📓 My Notes: Activity 4 — Series vs. Parallel Circuits

Copy these notes into your science notebook!

Key Vocabulary

Series Circuit

A circuit where all components are connected in ONE single loop. Current has only one path to follow.

Parallel Circuit

A circuit where components are connected in MULTIPLE branches. Current has more than one path to follow.

Load

Any component in a circuit that uses electrical energy (e.g., a light bulb, motor, or buzzer).

Short Circuit

When current takes an unintended shortcut path, bypassing the load — this can cause overheating!

Series vs. Parallel — Side by Side

Series Circuit	Parallel Circuit
One path for current	Multiple paths for current
If one bulb goes out, ALL bulbs go out	If one bulb goes out, the OTHERS stay on
Adding more bulbs makes them ALL dimmer	Adding more bulbs does NOT make them dimmer
Voltage is shared between components	Each branch gets the FULL voltage

Important Concepts to Copy

> In a series circuit, the same current flows through every component.

> In a parallel circuit, the current splits — each branch carries only part of the total current.

> The total current increases as more branches are added, because each branch provides another path.

> Most home electrical systems use PARALLEL circuits so that one broken device doesn't shut everything off.

✏️ **Draw It!**

"In your notebook, draw TWO circuits side by side. On the left, draw a SERIES circuit with a battery and 2 light bulbs in one loop. On the right, draw a PARALLEL circuit with a battery and 2 light bulbs on separate branches. Label both diagrams."

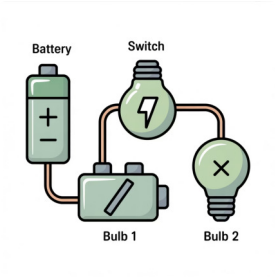
My Series & Parallel Diagrams

Detailed Teaching Guide: Activity 5 – Design Challenge: Building from Schematics

Welcome to our final circuit challenge! Today, we're going to put all your knowledge of circuits to the test. You'll be given schematic diagrams, and your task will be to build these circuits in the simulator, predict their behavior, and then test them to see if your predictions hold true. This activity will truly solidify your understanding of open versus closed circuits and how each component plays a role.

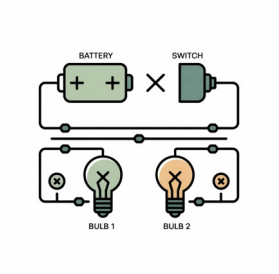
📐 Challenge Schematics

Challenge 1: Series Circuit



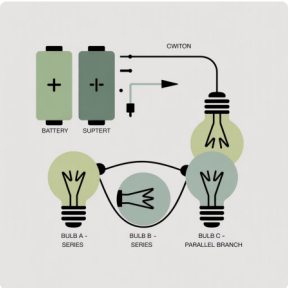
Battery + Switch + 2 Bulbs in series. One path for current.

Challenge 2: Parallel Circuit



Battery + Switch + 2 Bulbs in parallel. Multiple paths for current.

Challenge 3: Combination Circuit



Battery + Switch + Bulb A (series) + Bulbs B & C (parallel branch).

Teacher Says:

"Alright, future electrical engineers! We've learned about series, parallel, voltage, current, and resistance. Now, it's time to apply that to real-world circuit designs – well, simulated ones for now! For each challenge, I'll show you a schematic. Your mission, should you choose to accept it, is to:

1. Carefully interpret the schematic.
2. Build the circuit in your simulator.
3. Predict what will happen (e.g., will bulbs light up? Which ones? How brightly? What will happen if a wire is disconnected?).
4. Test your circuit and compare your observations with your predictions.
5. Explain your results, focusing on why certain parts are open or closed."

"Let's start with a foundational one. Here is your first schematic → a simple circuit with a battery, a switch, and two light bulbs connected in series. Build it, predict the bulb behavior, and tell me what happens if you open the switch."

Class Does:

- Open their circuit simulators and carefully construct the series circuit diagrammed.
- Discuss predictions for brightness and the effect of the switch, noting that opening the switch will create an open circuit, stopping current flow to both bulbs.
- Close the switch, observe the bulbs, and then open the switch to verify the prediction.
- Confirm that both bulbs light up when the circuit is closed and go out when the switch creates an open circuit.

Teacher Says:

"Excellent! Now for challenge number two. This schematic is a bit different. You'll see a battery, a switch, and two light bulbs, but this time they are connected in parallel. Build this, predict how the bulbs will behave compared to the series circuit, and then tell me what happens if you disconnect one of the bulbs while the circuit is closed."

Class Does:

- Disassemble the previous circuit and build the parallel circuit from the new schematic.
- Predict that the bulbs will be brighter than in the series circuit. They also anticipate that if one bulb is disconnected, the other will remain lit because the current has an alternative closed path.
- Test the circuit, observe brightness, and then simulate disconnecting one bulb.
- Confirm that the bulbs are bright and that removing one bulb does not affect the other, demonstrating independent closed paths.

Teacher Says:

"Fantastic observation on the independent paths! Now for the ultimate test: a combination circuit. Look at this next schematic carefully. It combines both series and parallel elements. You'll see a battery, a switch, and three bulbs: two in parallel with each other, and that parallel combination then connected in series with the third bulb and the switch. This is where your understanding of open and closed paths really shines. Predict what happens to ALL bulbs if:

1. The main switch is open.
2. One of the bulbs in the parallel section burns out (becomes an open circuit).
3. The bulb in the series section burns out.

Build it, test it, and then be ready to explain every outcome!"

Class Does:

- Construct the complex series-parallel circuit according to the schematic.
- Collaborate to make detailed predictions for each scenario, applying their knowledge of current paths in both series and parallel sections. They discuss how an open circuit in one part can affect the entire circuit or just specific branches.
- Conduct tests for all three scenarios, carefully observing the behavior of each bulb.
- Provide comprehensive explanations, correctly identifying when the entire circuit becomes open, or when only a specific branch is affected, linking their observations back to the concepts of open and closed circuits.

Teacher Says:

"Absolutely brilliant work, everyone! You've successfully navigated complex schematics, made accurate predictions, and explained the outcomes using the fundamental principles of open and closed circuits. This ability to read schematics and predict circuit behavior is a critical skill for anyone working with electronics. You've truly mastered the foundational concepts of electricity. Give yourselves a round of applause!"

📓 My Notes: Activity 5 — Reading & Building from Schematics

Subtitle: "Copy these notes into your science notebook!"

Key Vocabulary

Schematic Diagram
A drawing that uses standard symbols to represent the parts of a circuit. Engineers and electricians use these to plan and build circuits.

Symbol
A simple picture or shape that stands for a real circuit component (e.g., a long and short line = battery).

Switch
A component that opens or closes a circuit to control the flow of current.

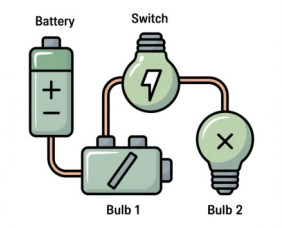
Resistor
A component that reduces the flow of current in a circuit. Shown as a zigzag line in a schematic.

Common Schematic Symbols to Copy

1	**Battery** Two lines — one long (positive +) and one short (negative –)
2	**Light Bulb** A circle with an X inside
3	**Switch (open)** A line with a gap and a diagonal line (like an open door)
4	**Switch (closed)** A straight, connected line (like a closed door)
5	**Resistor** A zigzag line
6	**Wire** A straight line connecting components

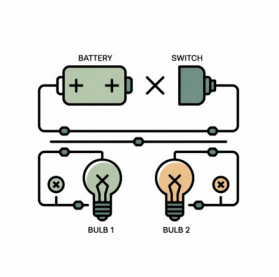
📐 The 3 Challenge Schematics — Study These!

Challenge 1: Series Circuit



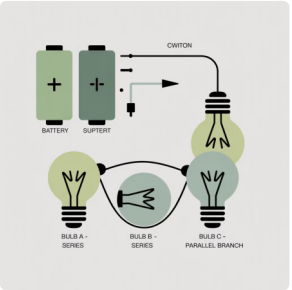
One loop. If the switch opens, BOTH bulbs go out.

Challenge 2: Parallel Circuit



Two branches. If one bulb goes out, the OTHER stays on.

Challenge 3: Combination Circuit



Bulbs B & C are parallel. Bulb A and the switch are in series with the whole parallel section.

Important Concepts to Copy

- Always read a schematic before building — trace the path of current from the positive terminal of the battery, through each component, and back to the negative terminal.
- A schematic is like a map for a circuit. The symbols are the same no matter what country you're in!
- If a switch is OPEN in a schematic, the circuit is broken and current cannot flow.
- If a switch is CLOSED in a schematic, the circuit is complete and current CAN flow.

✏️ **Draw It!**

"In your notebook, draw the schematic symbols for: a battery, a light bulb, an open switch, a closed switch, and a resistor. Label each one. Then connect them all into one complete circuit schematic."

My Schematic Symbols & Circuit

Teacher-Student Q&A: Test Your Circuit Smarts!

Time for a fun, interactive quiz to consolidate our understanding of electric circuits! Here are some key questions and answers from our recent explorations, making it easy to test your knowledge.

1	2
Question: What's the first thing you need to ensure for a bulb to light up in the PhET simulator?	Answer: The circuit must be **closed** (no gaps) for electrons to flow.
3	**4**
Question: What happens if a circuit is **open**?	Answer: Nothing lights up; the current cannot flow.
5	**6**
Question: What kind of material allows electricity to flow between the battery and the bulb?	Answer: A **conductor** (like copper wire). An **insulator** (like rubber) blocks the flow.
7	**8**
Question: What is the difference between what a voltmeter measures and what an ammeter measures?	Answer: A voltmeter measures **voltage** (the "push" of electricity), and an ammeter measures **current** (the flow of electrons).
9	**10**
Question: How do bulbs behave differently in **series** versus **parallel** circuits if one burns out?	Answer: In **series**, if one bulb goes out, all go out. In **parallel**, the others stay lit because there are multiple pathways.
11	**12**
Question: What PhET simulator feature helps visualize the flow of electricity?	Answer: The "blue balls" (moving dots) visually represent the **current** and its speed.

Conclusion: From Watching to Doing — Empowering Students to Discover Electricity

This lesson completes a powerful learning arc: from Bill Nye's entertaining explanations and BBC's visual storytelling, to the hands-on virtual laboratory of the PhET simulator. Students don't just know about electricity — they've experienced it, tested it, and made it their own.

🎬 Watched	🔬 Explored	💡 Discovered
Bill Nye & BBC introduced concepts with energy and clarity — sparking curiosity and building vocabulary.	PhET transformed abstract ideas into visible, interactive experiments — students built, broke, and rebuilt circuits safely.	Through reflection and assessment, students connected theory to reality — ready for deeper study of electric circuits.

The PhET simulation is **completely free** and requires no installation — bookmark phet.colorado.edu for every future lesson on circuits.

ISTE Standards Alignment

This electricity and circuit lesson, integrating the PhET simulator with foundational video content, directly addresses several key ISTE Student Standards. By fostering active engagement and critical thinking through technology, students are empowered to become responsible and innovative learners in the digital age.

1.1 Empowered Learner

Students leverage technology to take an active role in choosing, achieving, and demonstrating competency in their learning goals. The PhET simulation allows them to explore electrical concepts at their own pace, experiment freely, and immediately observe the outcomes of their decisions, thus deepening their understanding and ownership of the learning process.

1.3 Knowledge Constructor

Students critically curate information and use digital tools to construct knowledge, make meaningful learning experiences for themselves and others. By building circuits, testing materials, and observing the flow of electricity, students are actively constructing their understanding of complex scientific principles, moving beyond passive information reception.

1.4 Innovative Designer

Students use a variety of technologies within a design process to identify and solve problems by creating new, useful, or imaginative solutions. The challenge extensions, such as designing a circuit to achieve specific outcomes, encourage students to apply their knowledge creatively and iterate on their designs within the simulator environment.

1.5 Computational Thinker

Students develop and employ strategies for understanding and solving problems in ways that leverage the power of technological methods to develop and test solutions. The structured activities require students to predict, test, and analyze results, fostering computational thinking skills as they work through the logical steps of circuit design and behavior.

Next Lesson: Snap Circuits

Having mastered the fundamentals of electricity and circuits in the virtual laboratory, students are now ready to translate their digital understanding into tangible, real-world construction. The next phase of our journey introduces Snap Circuits, a hands-on system that allows students to physically build and experiment with electronic circuits using snap-together components.

This transition from the PhET simulation to physical Snap Circuits provides a crucial bridge between abstract concepts and practical application. Students will reinforce their knowledge of series and parallel circuits, voltage, current, and resistance by directly manipulating components. It's an opportunity to see how the theoretical principles they've explored digitally manifest in physical systems, deepening their intuitive grasp of electrical engineering.

Physical Assembly

Students will physically connect components, assembling circuits that light bulbs, make sounds, and power motors, observing immediate, real-world results.

Hands-On Troubleshooting

Building physical circuits often introduces new challenges, encouraging students to diagnose and fix problems, just like real engineers.

Creative Design

With a diverse set of components, students will be challenged to design and innovate their own circuits, bringing their ideas to life.

Teacher Says:

Fantastic work, everyone! You've all done an incredible job navigating the virtual world of electricity and circuits with the PhET simulator. You've learned about current, voltage, resistance, and how these fundamental principles make our modern world light up and buzz with energy. It's truly impressive how you've constructed knowledge and problem-solved like real engineers in the digital space.

Now, get ready for the next exciting step! In our next lesson, we're taking everything you've learned and bringing it to life with Snap Circuits. You'll get to physically build, test, and even troubleshoot your own electronic creations. It's going to be an amazing opportunity to see your virtual understanding transform into tangible, working circuits. I can't wait to see what innovative designs you'll come up with!